Precision Farming on Sugarcane: Drone-Based Disease Detection Using YOLOv8 Neural Models

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***Abstract*—Precision agriculture is being revolutionized by the use of UAVs and AI, enabling more efficient and sustainable crop monitoring. This study presents a drone-based solution for real-time detection of sugarcane diseases such as Rust, Red Rot, Mosaic, and Yellow Leaf. A custom quadcopter, outfitted with a high-resolution camera and Raspberry Pi 4, is used to capture aerial imagery. The onboard YOLOv8 model processes images in real time, with data stored locally on an SD card for further evaluation. The paper covers the complete system setup, including hardware components, neural network deployment, and the end-to-end workflow—from image capture to decision support. This integrated approach supports early intervention, better yield outcomes, and cost-effective disease management in sugarcane farming.**

***Index Terms*—Precision agriculture, sugarcane, disease detection, unmanned aerial vehicles, YOLOv8, Raspberry Pi, deep learning.**

I. Introduction

Precision agriculture uses data-driven technologies and UAVs integrated with AI/ML to optimize farm management, enhancing efficiency, sustainability, and decision-making. Conventional techniques tend to be time-consuming and lack precision when applied to large-scale area monitoring. Sugarcane, a globally important crop, faces significant challenges from diseases, causing up to 40 percent annual losses, making conventional field monitoring difficult. The critical global need for food security and reduced crop losses underscores the urgency for scalable agricultural solutions, which AI-powered drones directly address by optimizing resource use and enabling proactive disease intervention. Early and accurate disease detection is paramount to prevent widespread impact, as visible symptoms typically signify advanced infection. Timely intervention leads to improved yields and reduced operational costs. AI-based UAV systems achieve over 90 percent accuracy in crop disease detection, making them indispensable. AI's predictive analytics can identify disease indicators even before physical symptoms appear by analyzing subtle plant changes. This proactive capability fundamentally shifts disease management from reactive to preventative, maximizing intervention effectiveness, minimizing broad chemical use, and promoting environmental health UAVs provide high-resolution aerial imagery for early disease identification across vast fields, significantly reducing labor and improving accuracy. AI and ML models, including CNNs and SVMs, are crucial for crop disease identification and yield forecasting. YOLO (You Only Look Once) is a prominent real-time object detection model adept at rapidly identifying disease-affected areas in drone imagery. This synergistic integration of UAVs for large-scale data acquisition and AI for sophisticated data interpretation overcomes manual inspection limitations. The combined deployment enables precise, data-driven interventions, revolutionizing crop health monitoring with automated, real-time insights and decision support.[1],[2],[3],[4],[5],[6] This research addresses the critical need for an efficient, real-time, and cost-effective system for detecting sugarcane diseases (Rust, Redrot, Mosaic, and yellow leaves). This paper proposes a precision farming system featuring a custom-built quadcopter, an on-board Raspberry Pi 4 for localized processing, and YOLOv8 neural models for automated disease identification. Key contributions include:

- A comprehensive system architecture for drone-based sugarcane disease detection.
- Detailed specifications and integration strategies for the drone's hardware components—including the quadcopter frame, flight controller (e.g., Pixhawk or Matek), power distribution system (Li-Po battery with ESCs), on-board computer (such as the Raspberry Pi 4 Model B), and vision system (e.g., Insta360 Ace Pro camera)—are discussed to ensure seamless communication, power management, and real-time data processing for autonomous agricultural navigation and plant health monitoring.
- An analysis of YOLOv8 model performance on a resource-constrained edge device (Raspberry Pi 4) and optimization strategies.
- A detailed, step-by-step flowchart illustrating the entire data processing and hardware implementation workflow.

II. Related Works

A. SUGARCANE DISEASES: CHARACTERISTICS AND VISUAL SYMPTOMS

Precisely diagnosing sugarcane diseases is essential for implementing successful control strategies. The proposed system focuses on detecting four specific conditions: Rust, Redrot, Mosaic, and general yellow leaf symptoms. Understanding the distinct visual characteristics of each is crucial for training and validating the neural detection model.

Fig. 1. Rust (Puccinia melanocephela / Puccinia kuehnii).

Fig. 2. Red Rot (Glomerella tucumanensis)

Fig 1: Rust is a major foliar disease in sugarcane, triggered by the parasitic fungi Puccinia melanocephala (commonly linked to brown rust) and Puccinia kuehnii (responsible for orange rust). The infection mostly targets sugarcane leaves, where spore-generating pustules tend to appear, especially on the underside. In the case of P. melanocephala, these pustules produce characteristic orange-brown spores. The disease thrives under specific weather conditions—cool nights accompanied by dew, followed by warm and sunny days typical of dry seasons. Wind and rain-splash are the primary means through which rust spores disperse quickly across fields, enabling fast disease transmission. Early signs appear as narrow, yellowish streaks on both sides of the leaf, which gradually extend and shift in color to shades of brown, orange-brown, or reddish-brown, usually ranging from 2 to 10 mm in length. In highly susceptible varieties, numerous pustules can coalesce, forming large necrotic areas that may lead to premature leaf death, making severely infested crops appear brown or reddish from a distance. The distinct visual characteristics and environmental preferences of these rust types necessitate a highly robust detection model capable of accurately distinguishing similar diseases and adapting across varying conditions. This specificity suggests that integrating environmental metadata could refine detection accuracy and enable predictive modeling [7], [8].

Fig 2: Red Rot is one of the most destructive fungal diseases in sugarcane, caused by Glomerella tucumanensis (asexual stage: Colletotrichum falcatum). It typically begins with external foliar symptoms, such as a color transition in the third or fourth leaf, shifting from green to orange and then yellow, followed by bottom-up drying of leaves. Reddish lesions may form on the underside of the leaf midrib if fungal spores enter through the sheath. These symptoms manifest externally about 16–21 days post-infection, and the entire stalk can desiccate within 10 additional days if left untreated.The more conclusive diagnosis, however, relies on internal symptoms. Splitting an infected stalk reveals reddish-brown longitudinal streaks interspersed with white patches, sometimes accompanied by a blackish-brown liquid and an alcoholic odor from decaying pith tissues. In advanced stages, black fruiting bodies (acervuli) can form along the rind and nodes of the stalk. From a UAV-based precision agriculture standpoint, RGB imaging with consumer-grade cameras such as the Insta360 Ace Pro enables the detection of external foliar symptoms—like discoloration or leaf drying—particularly when visible from the canopy. However, the diagnostically significant internal symptoms of red rot are not detectable with RGB imaging, which restricts the scope of detection for this disease.This limitation underscores the need for field verification (groundtruthing) or integration of advanced sensing modalities in future iterations. While near-infrared (NIR), thermal, or hyperspectral sensors have been shown to detect internal plant stress these were not used in our current setup. The Insta360 Ace Pro, although efficient for high-resolution RGB imaging and post-processing, does not support infrared imaging, making it unsuitable for internal stress detection in red rot cases.In this study, once early-stage leaf symptoms are detected via UAV, they serve as triggers for targeted manual inspection, enabling a cost-effective disease surveillance workflow. Combining drone-based de-

tection with on-ground validation and progressive dataset development strengthens the overall reliability of the monitoring system.[2],[4],[5],[9],[10],[27],[28],[34],[40]

Fig. 3. Mosaic (Sugarcane Mosaic Virus - SCMV)

Fig 3: Mosaic disease in sugarcane, caused by the Sugarcane Mosaic Virus (SCMV), is identified by a distinctive mosaic pattern comprising alternating chlorotic (yellow) and green stripes on the leaf surface, often more pronounced in younger foliage. These yellowish areas become especially visible when affected leaves are held against the light, highlighting the uneven discoloration. As the disease progresses, expanded chlorotic patches, necrotic lesions in internodes, and stunted chlorotic plants may appear. Symptom severity often increases under cooler climatic conditions such as late summer, autumn, and winter. Due to seasonal and phenological variability in symptom expression, constructing a robust deep learning model for disease recognition requires a training dataset that captures diverse growth stages, lighting conditions, and Disease Name Causal Agent Primary Affected Plant Part Key Visual Symptoms (External) Distinguishing Features / Notes Rust Puccinia melanocephela / Puccinia kuehnii Leaves Small yellowish spots turning brown/orange-brown/red-brown; chlorotic halo; pustules (spores) on lower leaf surface; premature leaf death. Two types (common/orange) with subtle spore color differences; environmental preferences (wet/dry season). Redrot Glomerella tucumanensis Leaves, Stalk/Internal Leaves green to orange to yellow, drying bottom-up; reddish spots on leaf midrib (backside). Internal: reddish pith with white tinges; alcohol odor; acervuli on rind/nodes (not externally visible); delayed external symptom appearance. Mosaic Sugarcane Mosaic Virus (SCMV) Leaves Blotchy/streaky patterns of yellow and green on leaves (mosaic pattern); yellowish strips alternating with green; stunting. More prominent on younger foliage; noticeable in cooler temperatures; mechanically/insect transmitted. Yellow Leaf Disease Sugarcane Yellow Leaf Virus (SCYLV) Leaves Yellowing of leaf midrib on underside (leaves 3–6), expanding to blade; general yellowing from distance; stunted growth; bunchy appearance. Yellowing can be non-specific (stress, etc.); most noticeable in mature cane; aphid transmitted. environmental scenarios. Raspberry Pi 4 Model B, when combined with imaging modules like the Camera Module 3 or Insta360, offers an edge-computing platform suitable for deploying convolutional neural networks (CNNs) or YOLOv8 object detection models directly onboard UAVs. This eliminates reliance on ground stations and reduces latency in decision-making during flight operations.[1],[7],[10],[11],[12],[16],[23],[24],[32],[33],[34]

Fig. 4. Yellow Leaf Disease (Sugarcane Yellow Leaf Virus – SCYLV)

Fig 4: Yellow Leaf Disease (YLD), caused by the Sugarcane Yellow Leaf Virus (SCYLV), is a viral infection that significantly affects sugarcane yield and quality. Early symptoms typically manifest as yellowing of the midrib on the underside of the leaf, beginning around the third to sixth leaf from the top of the plant. As the disease progresses, the yellowing spreads laterally into the leaf blade, leading to an overall chlorotic appearance that becomes increasingly visible at the field level, especially during the post-monsoon period from October to March. At later stages of infection, affected plants show signs such as reduced growth, spindle desiccation, leaf reddening, and a compact, bunchy top—indicative of severe physiological stress caused by the disease. For drone-based remote sensing applications using RGB cameras like the Insta360, such midrib yellowing and leaf discoloration can be captured from an aerial perspective, especially at later stages when the symptoms intensify. However, a major challenge lies in the visual overlap between YLD symptoms and non-disease-related stressors such as nutrient deficiencies, insect damage, or drought, which also cause generalized leaf yellowing [10],[13],[14],[27],[40].

TABLE I
KEY VISUAL SYMPTOMS OF SUGARCANE DISEASES

Disease	Causal Agent	Primary Affected Part	Key Visual Symptoms
Rust	*Puccinia melanocephela / kuehnii*	Leaves	Small yellowish spots; brown/orange pustules; premature leaf death.
Redrot	*Glomerella tucumanensis*	Leaves, Internal	Leaf discoloration; internal reddish-brown streaks with white patches.
Mosaic	*Sugarcane Mosaic Virus (SCMV)*	Leaves	Yellow-green mosaic pattern; chlorotic streaks; stunting.
YLD	*Sugarcane Yellow Leaf Virus (SCYLV)*	Leaves	Yellowing of leaf midrib and blade; stunted growth; bunchy top.

III. METHODOLOGY

A. Quadcopter Frame and Payload Capacity: The foundation of the precision farming system is a custom-developed quadcopter platform. A critical design parameter for this drone is its ability to withstand a payload of 3 kg. This substantial payload capacity is essential for accommodating the necessary on-board components, including the Raspberry Pi 4, the camera module, and other potential sensors required for comprehensive crop monitoring. Agricultural drone frames are typically engineered for robustness and durability, often constructed from materials like carbon fiber, enabling them to carry a wide range of agricultural sensors and equipment effectively. he designated 3 kg payload plays a key role in shaping decisions related to motor choice, battery specifications, and the drone's overall size and structure. While this capacity provides ample room for integrating powerful on-board processing hardware, it also directly impacts the drone's power consumption and, consequently, its flight time. This presents a significant design trade-off, as limited UAV battery life is a known constraint in precision agriculture. Optimizing the balance between payload capability and flight endurance is a key challenge in such aerial platforms.[2],[15].

B. Flight Control System: Pixhawk Cube+ Integration: The quadcopter's flight control is managed by a Pixhawk Cube+ flight controller, which is integrated with the drone's motors. The Pixhawk platform is widely recognized in the drone community for its open-source nature, high dependability, and robust performance. The Cube+ variant boasts a triple redundancy power system and a sophisticated triple IMU (Inertial Measurement Unit) system, incorporating sensors such as ICM42688, ICM20948, ICM20649, ICM45686, and MS5611x barometers. These features ensure exceptional stability and accuracy during flight, even in challenging weather conditions, resulting in clear, shake-free shots. This stability is particularly vital for acquiring clear, high-quality images, which are indispensable for accurate deep learning inference. The Pixhawk Cube+ is highly versatile, compatible with multiple vehicle configurations and communication standards, which allows it to integrate seamlessly with different drone systems. Its precision control capabilities are crucial for maintaining stable flight paths during data acquisition over agricultural fields, ensuring consistent image quality for the disease detection model.[16],[17].

C. Power System: 6200mAh LiPo Battery: The quadcopter is powered by a 6200mAh LiPo battery with a discharge capacity of up to 35C. This battery capacity is selected to provide sufficient power for the drone's motors, the Raspberry Pi 4, and the camera system, while supporting the 3 kg payload. A 6S 22.2V 6200mAh LiPo battery pack is a suitable choice for medium to large-sized aerial photography equipment, known for packing substantial power in a compact form factor, which is essential for heavy-lift aerial photography rigs. LiPo batteries, such as those from Tattu, are known for providing high quality and reliable power for multirotors.The drone's flight duration largely depends on its battery capacity, energy usage, and operational efficiency. Given the 3kg payload, the drone's power draw will be substantial. For instance, if a conservative estimate of 170 W/kg is used for power required to lift equipment, and assuming a total drone weight (including battery and payload) of around 5-6 kg, the average amp draw would be significant. For a 6S (22.2V) 6200mAh battery, with a typical 80 percent discharge rule to prevent damage, the effective capacity is 4960mAh. While precise flight time calculations require detailed motor and propeller specifications, this battery configuration suggests a flight duration appropriate for targeted scouting missions rather than extensive, long-range mapping. For aerial photography work, where minimal maneuvering occurs, flight times can be estimated at approximately 75 percent of the theoretical hover time. This implies that mission planning must account for battery limitations, potentially requiring multiple flights for larger agricultural areas.[18],[19],[20],[21].

D. Raspberry Pi 4 as the Edge Computing Platform: The

Raspberry Pi 4 is used as the on-board microprocessor, enabling real-time edge computing for image analysis on the drone. Cost-effective and portable, it supports sensor control and data processing in-field, ideal for precision agriculture. With sufficient RAM and processing power, it's suitable for deploying YOLOv8 models, reducing latency by avoiding raw image transfer to cloud/ground stations.[1],[22].

E. High-Resolution Camera Module: A Insta360 Ace Pro camera is mounted on the quadcopter and connected to the Raspberry Pi 4 via USB as a UVC webcam. It provides 4K resolution, high dynamic range, and image stabilization, enabling detailed, stable footage suitable for real-time aerial disease monitoring. The wide-angle lens enhances field coverage, while its video output supports fast frame capture needed for object detection. Although multispectral and hyperspectral cameras offer deeper physiological data, a high-resolution RGB feed from the GoPro combined with YOLOv8 provides sufficient detail to detect visible disease symptoms like lesions, yellowing, or mosaic patterns under natural lighting conditions.[3]

IV. NEURAL MODEL: YOLOv8 FOR DISEASE DETECTION

A. YOLOv8 Model Architecture and Suitability:

YOLOv8 (You Only Look Once version 8) is a state-of-the-art object detection model known for real-time performance. It performs object recognition and bounding box regression in a single forward pass, enabling image processing at standard frame rates depending on configuration.. By dividing images into grids and predicting bounding boxes with class probabilities, it effectively detects multiple objects across regions. In agricultural contexts, customized YOLOv8 models have achieved high precision and recall for plant disease detection . Its adaptability to input resolutions and multitasking (detection, classification, segmentation) makes it suitable for drone-based monitoring [3],[29],[30].

B. Training and Dataset Considerations:

Training a YOLOv8 model for sugarcane disease detection requires a large, high-quality labeled dataset. Studies have used 4K drone imagery to collect data on diseases like rust and red rot, enhancing datasets via augmentation methods such as flipping and gamma adjustment to improve model robustness. Image collection and annotation are essential for custom training, With datasets from platforms like Roboflow serving as key resources. The variability in rust types, Mosaic symptom expression by plant age and environment, and the general yellowing caused by stress, insects, or water deficiency highlight the need for a diverse, representative dataset [8], [14]. Accurate model training must enable YOLOv8 to discern SCYLV-specific yellowing from non-disease causes, potentially requiring nuanced feature learning or multi-label classification.[3],[22],[33],[34].

C. Performance on Raspberry Pi 4 and Optimization:

YOLOv8 on resource-constrained edge devices like the Raspberry Pi 4 involves key performance challenges. Despite optimizations, real-time processing is limited by hardware constraints, as the Raspberry Pi 4 processor struggles to maintain high frame rates (e.g., 30 FPS) [22], [35]. Even with INT8 quantization, the more powerful Raspberry Pi 5 has shown difficulty in achieving real-time performance, unlike devices such as Jetson Orin [32]. In this work, inference performance on the Raspberry Pi 4 was optimized by leveraging default image resolution settings and minimizing post-processing overhead. To mitigate over-detection caused by bright regions such as the sky or lighting glare, a green-boundary filtering technique was employed. The algorithm isolates the leaf by detecting continuous green pixel regions using HSV color thresholding. This ensures that only biologically relevant areas, typically sugarcane leaves are analyzed for disease detection. Non-leaf regions are blurred out, improving both detection accuracy and visual clarity.

Fig. 5. Real-Time Detection

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D. Real-Time Visualization with Green-Region Isolation:

To enhance disease detection accuracy and suppress false positives in visually noisy environments (e.g., bright skies

or human hands), a green-boundary based leaf isolation method was integrated into the YOLOv8 detection pipeline. As shown in Fig. 5, the algorithm uses HSV color space thresholding to segment continuous green regions, which correspond to sugarcane leaves, while blurring out irrelevant background. The result is a visually enhanced frame where only leaf regions remain sharp and subject to detection. This not only improves the clarity of detected diseases like rust and mosaic but also ensures the model focuses on biologically significant areas, reducing inference errors due to lighting or unrelated yellowish hues

V. DATA WORKFLOW AND IMPLEMENTATION

The operational workflow of the drone-based precision farming system involves a sequence of steps, from image acquisition to on-board processing, data storage, and subsequent analysis. This integrated approach ensures efficient and actionable insights for sugarcane disease management.

A. System Workflow:

The overall workflow for the drone-based disease detection system is depicted in Fig

B. Workflow Description:

- Drone Deployment Over the Fields: A quadcopter with a Pixhawk Cube+ flight controller was flown autonomously or semi-autonomously over sugarcane fields for stable navigation and consistent image capture [4], [17]..
- Image Acquisition by Camera: A Insta 360 camera, mounted on the drone and operating in USB webcam mode, captured high-resolution canopy images during flight. This mode supports real-time streaming and simplifies integration over CSI modules [23]..
- Raw Image Data Streamed to Raspberry Pi 4: The USB-connected camera streams frames to the Raspberry Pi 4. While it bypasses CSI, USB mode offers adequate throughput for edge inference and requires no driver-level customization.
- Pre-processing on Raspberry Pi 4: Frames are resized (e.g., 416x416 or 640x640), normalized, and formatted to match YOLOv8 input requirements, optimizing the pipeline with minimal compute load.
- YOLOv8 Neural Model Inference (Disease Detection): Pre-processed images are analyzed by YOLOv8 on the Raspberry Pi 4, detecting Rust, Redrot, Mosaic, and yellow leaves [29]. Due to CPU limits, real-time (30 FPS) is not achievable [22], [35];optimizations included reduced post-processing, default resolution use, and false-positive filtering.
- Detection Output and Data Storage: YOLOv8 outputs bounding boxes, labels, and confidence scores, which are saved to the Raspberry Pi's SD card for offline access.
- Post-Processing and Optional Mapping: Although not implemented, stored detections can be exported to create disease hotspot maps, offering spatial insights and enabling farm-scale analytics [38].
- Decision Support and Monitoring:Aggregated results aid agronomists by visualizing disease severity and spread [38]. The system supports iterative deployment for tracking treatment outcomes and disease progression [41].
- Targeted Intervention (Future Scope): Future integration could include localized spraying guided by disease maps, using precision sprayers or RTK-enabled drones to minimize chemical use and environmental impact [6].
- Subsequent Drone Flights for Monitoring/Review: The cycle is iterative. Post-treatment drone surveys enable effectiveness review and adaptive disease management across crop stages [41].

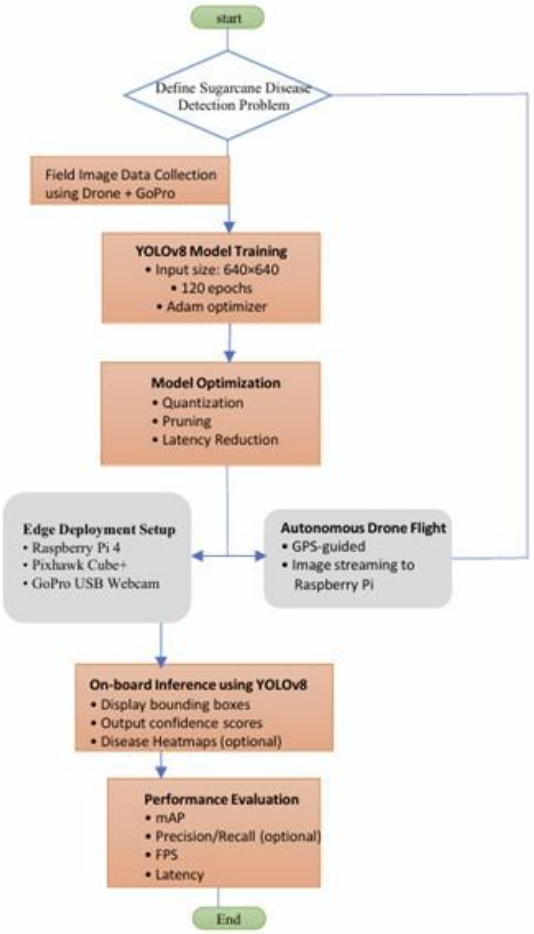

Fig. 6. Workflow

VI. CHALLENGES AND FUTURE DIRECTIONS

Despite the promise of AI-powered drones in sugarcane disease detection, several challenges persist, indicating key areas for future work.

Dataset Limitations: YOLOv8 and other deep learning models require large, high-quality, labeled datasets to ensure ac-

curacy [3]. Disease symptoms vary with crop age, soil, angle, moisture, weather, and sugarcane variety, affecting visual cues and spectral reflectance [5]. To improve generalization, future efforts must prioritize diverse data collection and standardized annotations.

Edge Processing Constraints: Real-time inference on edge devices like the Raspberry Pi 4 is limited by computational power. While YOLOv8 is efficient, maintaining high FPS without dedicated accelerators is challenging [32]. Future directions include model quantization, NPU integration, and custom AI chips for enhanced processing without excessive payload or power use [2].

Model Robustness and Environmental Adaptability: Generalizing across diverse field conditions, sugarcane varieties, and environmental noise (e.g., dust, rain, shadows) remains a key challenge [3]. Lighting issues like over/underexposure hinder symptom detection [3]. Future work will focus on building robust models through advanced image augmentation and exploring multi-modal AI using multispectral or thermal sensors for richer plant health insights [3].

Data Privacy and Federated Learning (FL): As agricultural data centralizes, privacy and security concerns grow. FL enables collaborative model training without sharing raw data, enhancing privacy and robustness [3]. However, it faces hurdles like data heterogeneity and high communication costs in rural areas [3]. Federated Learning (FL) allows multiple participants to train models jointly without exchanging raw data, thereby improving both privacy and system resilience which comes as a future aspect. [3]

Predictive Analytics for Proactive Intervention: Current systems detect visible symptoms; future efforts will shift toward predictive analytics using subtle metabolic, soil, and environmental indicators combined with historical data to forecast disease risks [3]. This would enable earlier, preventative actions, minimizing losses and optimizing inputs.

VII. CONCLUSION

Use of UAVs combined with YOLOv8 on edge devices (for example Raspberry Pi 4) represents a crucial step of precision agriculture in sugarcane cultivation. In this research an airborne autonomous compact system capable of real-time detection of major sugarcane diseases-Rust, Redrot, Mosaic, Yellow leaf using a GoPro ACE camera mounted on a quadcopter and processing will be presented. Data processing is performed locally and kept on the SD card, for real-time use and possible geospatial mapping.

By detecting diseases early, even before visible symptoms develop fully, this system provides an opportunity to be proactive in managing diseases, promoting crop health and combating use of pesticides. Though challenges in data quality, hardware constraints and model robustness persist, the system's capabilities are being strengthened by advances in sensors, lightweight AI and federated learning. Being low cost, scalable, and reconfigurable, such drone-based platforms are essential for addressing global food security challenges, optimizing agricultural productivity, and fostering sustainable farming practices in the sugarcane industry and beyond.

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